Datasheet

ALM-173-R1 Alarm Input & Relay Output Module



Part No.: ALM-173-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board)

Controller: RP2350A

Functional Overview

The ALM-173-R1 is a modular alarm input and relay output module designed for intrusion detection, supervision systems, and industrial annunciation panels. It provides 17 digital inputs and 3 SPDT relay outputs, with local user interface (4 buttons, 4 user LEDs). The module communicates over RS-485 as a Modbus RTU slave and is configured via USB-C Web Serial interface.

Board	Function
MCU Board	RP2350A controller, RS-485 communication, USB-C configuration interface, power conversion
Field Board	17 isolated digital inputs, isolated sensor power rails, Relays

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (W25Q32JV, 32 Mbit)
Power Input	24 V DC nominal
Input Protection	1 A fuse, reverse polarity diode, TVS surge suppression, EMI filtering
Main Logic Supply	Buck regulator 24 V → 5 V, 3.3 V LDO regulator
File system	LittleFS persistent configuration storage
Document revision	DS-ALM-173-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
Digital Inputs	17 × digital inputs, 5 V DC discrete inputs
Max Input Voltage	5 V DC continuous; input isolation 5.3 kVrms (SFH6156 optocoupler test voltage)
Relay outputs	3 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit)
Isolated +12 V Output	12 V isolated rail, 2 W; max 167 mA theoretical (≤ 150 mA recommended)
Isolated +5 V Output	5 V isolated rail, 1 W; max 200 mA theoretical (≤ 150 mA recommended)
User Interface	4 buttons; 27 LEDs: PWR, TX, RX; 4 user LEDs; 17 input LEDs (I1–I17); 3 relay LEDs (O.1–O.3)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C, 5 V logic, ESD protected
Modbus defaults	Address 3, 19200 baud, 8N1

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Typical module power consumption: 0.2–0.5 W (logic only).

Top Terminal Block (1–28)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	Gnd	DI1	DI1 isolated return
4	I.1	DI1	DI1 input
5	Gnd	DI2	DI2 isolated return
6	I.2	DI2	DI2 input
7	Gnd	DI3	DI3 isolated return
8	I.3	DI3	DI3 input
9	Gnd	DI4	DI4 isolated return
10	I.4	DI4	DI4 input
11	Gnd	DI5	DI5 isolated return
12	I.5	DI5	DI5 input
13	Gnd	DI6	DI6 isolated return
14	I.6	DI6	DI6 input
15	Gnd	DI7	DI7 isolated return
16	I.7	DI7	DI7 input
17	Gnd	DI8	DI8 isolated return
18	I.8	DI8	DI8 input

19	Gnd	DI9	DI9 isolated return
20	I.9	DI9	DI9 input
21	Gnd	DI10	DI10 isolated return
22	I.10	DI10	DI10 input
23	NC	RELAY1	Relay 1 normally closed
24	C	RELAY1	Relay 1 common
25	NO	RELAY1	Relay 1 normally open
26	NC	RELAY3	Relay 3 normally closed
27	C	RELAY3	Relay 3 common
28	NO	RELAY3	Relay 3 normally open

Bottom Terminal Block (1–28)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	+	PS_12V_1	Isolated 12 V rail 1 positive
2	-	PS_12V_1	Isolated 12 V rail 1 return
3	+	PS_12V_2	Isolated 12 V rail 2 positive
4	-	PS_12V_2	Isolated 12 V rail 2 return
5	+	PS_5V_1	Isolated 5 V rail 1 positive
6	-	PS_5V_1	Isolated 5 V rail 1 return
7	+	PS_5V_2	Isolated 5 V rail 2 positive
8	-	PS_5V_2	Isolated 5 V rail 2 return
9	COM	RS485	RS-485 signal reference
10	B	RS485	RS-485 data -
11	A	RS485	RS-485 data +

12	GND	DI11	DI11 isolated return
13	I.11	DI11	DI11 input
14	GND	DI12	DI12 isolated return
15	I.12	DI12	DI12 input
16	GND	DI13	DI13 isolated return
17	I.13	DI13	DI13 input
18	GND	DI14	DI14 isolated return
19	I.14	DI14	DI14 input
20	GND	DI15	DI15 isolated return
21	I.15	DI15	DI15 input
22	GND	DI16	DI16 isolated return
23	I.16	DI16	DI16 input
24	GND	DI17	DI17 isolated return
25	I.17	DI17	DI17 input
26	NO	RELAY2	Relay 2 normally open
27	C	RELAY2	Relay 2 common
28	NC	RELAY2	Relay 2 normally closed

Installation, Environmental & Mechanical

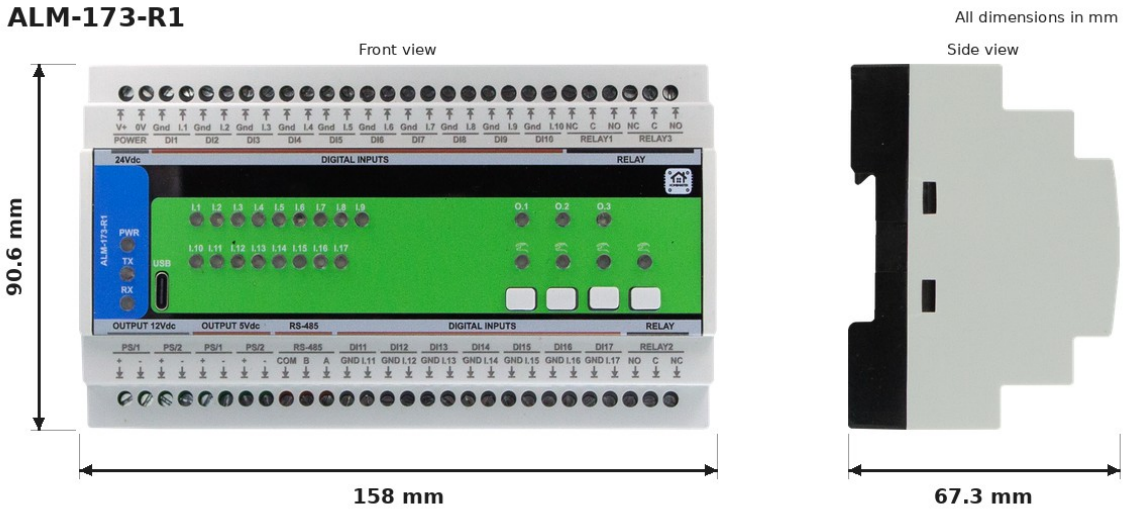
Specification	Details
Terminal Type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire Cross Section	0.2–2.5 mm^2 (AWG 24–12)
Tightening Torque	0.4–0.6 Nm

Specification	Details
Operating temperature	0 °C to +40 °C
Storage temperature	−10 °C to +55 °C
Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m
Pollution degree	2

Specification	Details
Dimensions	158 × 90.6 × 67.3 mm (L × W × H)
DIN width	9 modules (≈ 158 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm^2 (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	200 g
Gross weight	330 g
Pack size	230 × 140 × 87 mm (L × W × H)
Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm
Notes	All dimensions shown in millimeters unless stated otherwise

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

Compliance & Certifications

The ALM-173-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

Safety Notice

The power input and signal terminals are SELV circuits. Relay output terminals may switch hazardous voltage up to 250 VAC.

Adequate insulation and separation must be ensured during installation.Installation by qualified personnel only.

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